

Lecture Notes on Locally Linear Embedding (LLE)

1 Introduction to Locally Linear Embedding (LLE)

Locally Linear Embedding (LLE) is a powerful **nonlinear dimensionality reduction** technique based on the assumption that high-dimensional data lies on or near a low-dimensional manifold. Unlike linear methods like PCA, LLE preserves *local* structures, making it ideal for data that lies on complex, non-linear manifolds.

2 Goals of LLE

The main objective of LLE is to find a lower-dimensional representation of high-dimensional data such that:

- **Local relationships are preserved:** Points that are close in the high-dimensional space remain close in the low-dimensional space.
- **Nonlinear structures are captured:** Unlike PCA, LLE can unfold nonlinear manifolds effectively.

3 Mathematical Foundations of LLE

LLE consists of two main steps:

1. **Finding optimal weights** that reconstruct each data point as a linear combination of its neighbors in the high-dimensional space.
2. **Finding the low-dimensional embedding** that best preserves these local relationships using an eigenvalue decomposition.

3.1 Step 1: Finding the Optimal Weights

For each data point $\mathbf{x}^{(i)}$, LLE attempts to reconstruct it using its k -nearest neighbors. The reconstruction error is minimized by optimizing weights subject to certain constraints.

Identifying Nearest Neighbors

For each data point $\mathbf{x}^{(i)}$:

- Calculate the Euclidean distance to all other points.
- Select the k -nearest neighbors, denoted as the set $N(i)$.

Objective Function for Weights

Let $\hat{w}_{i,j}$ denote the weight that relates $\mathbf{x}^{(j)}$ (a neighbor of $\mathbf{x}^{(i)}$) to $\mathbf{x}^{(i)}$. LLE seeks weights that minimize the following reconstruction error for each data point:

$$\text{minimize} \quad \sum_{i=1}^m \left\| \mathbf{x}^{(i)} - \sum_{j \in N(i)} \hat{w}_{i,j} \mathbf{x}^{(j)} \right\|^2$$

subject to:

1. **Neighborhood constraint:** $\hat{w}_{i,j} = 0$ if $\mathbf{x}^{(j)}$ is not a neighbor of $\mathbf{x}^{(i)}$.
2. **Normalization constraint:** $\sum_{j \in N(i)} \hat{w}_{i,j} = 1$.

Solving the Constrained Least-Squares Problem

This optimization can be formulated using a Lagrange multiplier to enforce the normalization constraint. The Lagrange function for each point $\mathbf{x}^{(i)}$ is given by:

$$\mathcal{L}_i(\hat{w}_{i,j}, \lambda) = \left\| \mathbf{x}^{(i)} - \sum_{j \in N(i)} \hat{w}_{i,j} \mathbf{x}^{(j)} \right\|^2 + \lambda \left(\sum_{j \in N(i)} \hat{w}_{i,j} - 1 \right)$$

Setting the partial derivatives with respect to $\hat{w}_{i,j}$ and λ to zero yields a system of linear equations, which can be solved to obtain the optimal weights $\hat{w}_{i,j}$ for each point.

Constructing the Weight Matrix $\hat{W}$

After computing $\hat{w}_{i,j}$ for each data point, we construct the sparse weight matrix $\hat{W}$ where:

- $\hat{W}_{i,j} = \hat{w}_{i,j}$ if $j \in N(i)$,
- $\hat{W}_{i,j} = 0$ otherwise.

4 Step 2: Finding the Low-Dimensional Embedding

The second step of LLE involves finding a low-dimensional representation Z that preserves the local relationships encoded in $\hat{W}$.

4.1 Objective Function for Embeddings

Given the weight matrix $\hat{W}$, LLE seeks a low-dimensional representation $Z = [\mathbf{z}^{(1)}, \mathbf{z}^{(2)}, \dots, \mathbf{z}^{(m)}]^\top$, where each point $\mathbf{z}^{(i)}$ in this low-dimensional space satisfies:

$$\text{minimize} \quad \sum_{i=1}^m \left\| \mathbf{z}^{(i)} - \sum_{j=1}^m \hat{w}_{i,j} \mathbf{z}^{(j)} \right\|^2$$

This objective function ensures that each point $\mathbf{z}^{(i)}$ is approximated by its neighbors in the same way as in the high-dimensional space.

4.2 Constructing the Cost Matrix M

The objective function can be represented in matrix form:

$$\sum_{i=1}^m \left\| \mathbf{z}^{(i)} - \sum_{j=1}^m \hat{w}_{i,j} \mathbf{z}^{(j)} \right\|^2 = \text{tr}(Z^\top M Z)$$

where $M = (I - \hat{W})^\top (I - \hat{W})$.

Rationale Behind M

The matrix M quantifies the cost of preserving the neighborhood relationships defined by $\hat{W}$. Minimizing $\text{tr}(Z^\top M Z)$ ensures that points that were close in the high-dimensional space remain close in the low-dimensional space, according to the weights in $\hat{W}$.

4.3 Solving the Eigenvalue Problem

The minimization problem minimize $\text{tr}(Z^\top MZ)$ can be solved as an eigenvalue problem:

- Compute the eigenvalues and eigenvectors of M .
- Discard the eigenvector corresponding to the smallest eigenvalue (trivial solution. all points collapse to the same location).
- Use the next d smallest eigenvalues and their eigenvectors to construct Z , the low-dimensional embedding.

5 Computational Complexity of LLE

The computational complexity of LLE is as follows:

- Finding the k -nearest neighbors: $O(m \log(m)n \log(k))$
- Optimizing weights: $O(mnk^3)$
- Constructing the low-dimensional representation: $O(dm^2)$

The last term, $O(dm^2)$, makes LLE computationally expensive for large datasets.

6 Conclusion

Locally Linear Embedding is a powerful technique for dimensionality reduction, especially useful for data that lies on nonlinear manifolds. By focusing on preserving local relationships, LLE can unfold complex structures in the data, providing a meaningful lower-dimensional representation.